

NANDINI

Email: nandinigunda792@gmail.com

PH: 817-754-0342

Senior Data Engineer | Cloud & Big Data

PROFESSIONAL SUMMARY

- Over 10+ years of **data engineering expertise in SaaS start-up and enterprise environments**, designing and delivering scalable, reliable, and efficient data solutions that empower global strategy and product teams to make confident, data-driven decisions.
- Deep hands-on experience developing **batch and real-time data pipelines** leveraging **Airflow, Apache Spark (Scala & Python), Apache Kafka, and SQL**, with a proven ability to architect, optimize, and operate data platforms supporting fast-growing SaaS KPIs such as ARR, churn, CLTV, and cohort analytics.
- Demonstrated proficiency in **Python, SQL, Scala, and distributed computing paradigms**, building and optimizing ELT/ETL workflows, streaming pipelines, and curated data products that meet high SLAs and support self-service analytics for business stakeholders.
- Engineered and deployed high-performance ETL pipelines in Airflow that **processed 5TB+ of data** daily from diverse sources (application events, billing, user activity), ensuring schema validation, data quality, and lineage across Snowflake, BigQuery, and Redshift.
- Built **low-latency streaming architectures** using Kafka Streams and Spark Structured Streaming, enabling product teams to detect and act on user behavior patterns with sub-minute lag and **improving customer engagement** analytics.
- Architected reusable data frameworks, libraries, and standards in Python and Scala to streamline development, improve data quality, and enforce consistency across engineering teams, frequently **mentored** and upskilled peers in Airflow DAG design, Spark job tuning, and Kafka topic management.
- Led the successful migration of **50+ legacy batch ETL jobs** into orchestrated, cloud-native platforms using Airflow, Kubernetes, Terraform, and CI/CD pipelines — **reducing job failures by 40% and cutting operational overhead by \$100K annually**.
- Collaborated closely with cross-functional teams to design **cloud-first data architectures on AWS, GCP, and Azure**, leveraging managed services (Databricks, EMR, Dataflow, Pub/Sub, S3) to elastically scale compute and storage while optimizing for cost and performance.
- Skilled in data modeling for SaaS metrics, delivering high-quality schemas and datasets that empower analytics and business users with actionable insights via dashboards, APIs, and federated queries (Trino/Presto) all while maintaining strong data validation and monitoring practices (e.g., Great Expectations).
- Delivered self-service data platforms and tools that reduced business teams' dependency on engineering by providing curated, trusted, and documented data products with clear lineage and SLA adherence.
- Actively participated in Agile/Scrum processes, design and code reviews, and strategic planning to align technical solutions with evolving SaaS business needs, while fostering a culture of collaboration, accountability, and continuous improvement.
- Recognized for **domain expertise in Airflow, Spark, Trino, and Kafka**, and trusted to guide architectural decisions, teach and mentor peers, and drive innovation in data engineering practices — consistently delivering impactful solutions in fast-paced SaaS start-up environments.

TECHNICAL SKILLS:

Programming & Scripting: Python, SQL, Scala, Java, PySpark, PL/SQL, R, Bash, PowerShell, Groovy, Ruby

Big Data & Distributed Systems: Apache Spark (Core, SQL, Streaming, MLlib), Hadoop, MapReduce, HDFS, Hive, HBase, Sqoop, Pig, Kafka, Yarn, Apache Beam, Trino/Presto

Cloud Platforms:

AWS: EC2, S3, RDS, Lambda, Glue, DynamoDB, IAM, CloudFormation, CloudWatch, CloudTrail, SNS, SQS, ELB, VPC, EBS, Route 53, Code Commit, Amazon MQ, Amazon SimpleDB, Amazon ECS

Azure: Databricks, Data Lake, Data Factory (ADF), Blob Storage, Azure SQL DB, SQL Data Warehouse (SQLDWH), Cosmos DB, Azure Active Directory

GCP: BigQuery, Cloud Composer, Dataproc, Pub/Sub, Dataflow

Data Lakes & Warehousing: Delta Lake, Apache Hudi, Snowflake, Redshift, BigQuery, Azure Synapse, Hive, MongoDB, Cassandra, Teradata, Oracle, PostgreSQL, MySQL, Aurora DB, DB2, MS SQL Server

ETL & Orchestration: Apache Airflow, dbt, Azure Data Factory, AWS Glue, Informatica, Talend, SSIS, NiFi, Apache Beam

Data Governance & Quality: Great Expectations, Apache Atlas, Collibra, Data Lineage, MDM, Schema Registry, DBT Testing

CI/CD & DevOps Tools: Git, GitHub Actions, Jenkins, Azure DevOps, GitLab CI, Terraform, Docker, Kubernetes, Helm, Maven, Gradle, Ansible, Chef, Puppet, Bitbucket

Observability & Monitoring: Datadog, Prometheus, Grafana, CloudWatch, Azure Monitor, Monte Carlo

Machine Learning & Analytics: Spark MLlib, Scikit-learn, Time Series Analysis, A/B Testing, Clustering, NLP, Logistic Regression, Random Forest, SVM, Decision Trees, WEKA

BI & Visualization Tools: Tableau, Power BI, Looker

APIs & Data Formats: RESTful APIs, JSON, XML

Operating Systems & Environments: Linux (CentOS), Windows, Mac OS, OpenStack, Cloud Foundry

IDEs & Development Tools: Jupyter, PyCharm, VS Code, IntelliJ, SQL Developer, Eclipse, Sublime Text, Putty

Methodologies: Agile (Scrum/Kanban), CI/CD Pipelines, TDD, SDLC, UML

PROFESSIONAL EXPERIENCE

Sr. Data Engineer / Big Data Developer

Giant Eagle, Pittsburgh, PA

April 2023 to Present

Responsibilities:

- Partnered with product and business strategy teams at fast-growing SaaS start-ups to architect and deliver scalable, reliable data pipelines that empowered leadership to make confident, data-driven decisions across global markets.
- Designed and implemented robust ETL and ELT workflows leveraging **Airflow**, **Scala**, **Python**, and **SQL**, ingesting heterogeneous datasets (application usage, billing, customer engagement) into centralized data platforms to support analytics and forecasting.
- Built **sub-minute latency** streaming pipelines using Apache Kafka and Spark Structured Streaming, allowing real-time monitoring of key SaaS metrics like churn signals and trial conversion, **improving stakeholder response time by 30%**.
- Optimized Spark batch pipelines to **process 3TB+ of daily data**, reducing transformation runtime **by 40%** and improving cost efficiency for ML model inputs and dashboard refreshes.
- Designed and deployed reusable **data frameworks and libraries** in Python and Scala, standardizing data ingestion, validation, and transformation patterns, reducing developer onboarding time and improving code quality across the engineering team.
- Led migration of over **50 legacy ETL jobs** to AWS MWAA-hosted Airflow environment, **reducing failure rate by 45%**, improving SLA compliance, and enhancing system observability.
- Collaborated with cross-functional teams to design **data models and schemas** in AWS Redshift tailored for SaaS KPIs like ARR, CAC, CLTV, and cohort analysis, ensuring scalability and high query performance.
- Performed in-depth **data profiling and validation** that improved **accuracy of event-level metrics by 25%**, using automated QA tests and Great Expectations for schema enforcement and anomaly detection.
- Delivered **self-service data products**, including curated datasets, APIs, and metadata-rich tables, enabling business stakeholders to explore SaaS performance without engineering intervention.
- Provided expert-level guidance and mentorship on **Airflow DAG design**, **Spark optimizations**, and **Kafka topic management**, fostering a data-driven engineering culture and upskilling team members.
- Tuned Spark and Kafka workloads by optimizing partitioning, resource allocation, and checkpointing strategies, reducing job execution times and improving SLA adherence in multi-tenant SaaS environments.
- Worked within Agile teams to prioritize and deliver data engineering features, participating actively in design reviews, code reviews, and sprint planning, ensuring alignment with SaaS business priorities and timelines.
- Migrated legacy S3-based parquet workflows to **Apache Hudi** for efficient upsets and incremental processing, **reducing daily ETL runtime by 40%**.
- Integrated CI/CD pipelines (GitLab CI, Jenkins) and infrastructure-as-code tools (Terraform, Helm) for automated deployment of data pipelines and infrastructure in AWS, ensuring reproducibility and reducing manual intervention.
- Contributed to cloud-native architecture by deploying pipelines and data services on **AWS** using managed services like EMR, Redshift and EKS to scale data workloads elastically.
- Integrated **Apache Atlas** to enable column-level data lineage and tagging across ETL pipelines, improving traceability and accelerating impact analysis for schema changes.
- **Reduced infrastructure costs by over \$20K/quarter** by optimizing Spark configurations to run on AWS spot instances, decommissioning idle Kafka topics, and enforcing data retention policies.
- Authored technical documentation, playbooks, and onboarding guides for Airflow, Spark, and Kafka pipelines, ensuring operational readiness and knowledge sharing within the SaaS engineering team.

Environment: Airflow, Apache Spark (Scala, Python), Apache Kafka, Python, SQL, Scala, Snowflake, BigQuery, AWS, Redshift, S3, Lambda, EC2, IAM, CloudWatch, MWAA, EMR, EKS, Terraform, Helm, GitLab CI, Jenkins, Great

Expectations, Apache Atlas, Spark Structured Streaming, Cloud-based Data Warehouses, SaaS Metrics (ARR, CAC, CLTV), ETL/ELT Workflows, Real-time Streaming, Batch Processing

Data Engineer / Data Architect

Thomson Reuters Egan, MN

October 2021 to March 2023

Responsibilities:

- Designed and implemented highly scalable, fault-tolerant data pipelines and ETL/ELT workflows in a fast-paced SaaS environment, enabling real-time and batch analytics for global strategy and product teams. Leveraged **Python, SQL**, and distributed computing frameworks (**Spark, Scala**) to transform diverse datasets into actionable insights.
- Orchestrated complex workflows using **Apache Airflow**, building reusable DAGs to automate ingestion, transformation, and delivery of high-quality data across multiple environments, while incorporating monitoring and alerting mechanisms to ensure operational excellence.
- Delivered end-to-end streaming and batch data solutions on **Azure infrastructure**, consuming and enriching high-volume data from Kafka topics, and writing curated data products into **Azure Data Lake, Azure Synapse Analytics**, and **Azure Blob Storage** with stringent SLAs.
- Optimized Spark-based pipelines on Azure Databricks to **process 4TB+ of data daily**, implementing advanced partitioning and caching techniques that **reduced job latency by 35%** and **saved \$10K/month in computing costs**.
- Partnered with cross-functional teams to understand business needs, and translated them into robust, scalable data architectures — designing schemas, defining interfaces, and building reusable libraries that accelerated development for the wider engineering team.
- Architected and delivered real-time event-stream processing frameworks using Kafka Streams and Spark Streaming on **Azure Event Hubs**, enabling the business to detect and act on customer behaviors and platform signals in near real-time with high reliability.
- Developed self-service data products and Power BI dashboards used **by 100+ business stakeholders**, improving insight **delivery time by 50%** and reducing dependence on engineering.
- Created reusable validation and error-handling frameworks that increased pipeline reliability and **reduced data quality incidents by 60%**.
- Designed and maintained CI/CD pipelines for data workflows using Git, Terraform, Docker, and **Azure DevOps**, reducing deployment times and ensuring consistent, reproducible environments.
- Led migration of **40+ batch-heavy legacy jobs** to stream-first architecture using Kafka and Trino, resulting in **3x faster data delivery** while maintaining zero data loss.
- Worked extensively with SaaS product analytics data — clickstreams, engagement metrics, billing, and operational telemetry — to drive product strategy, improve customer experience, and identify growth opportunities.
- Mentored junior engineers and peers on best practices for distributed data processing, Python/Scala coding standards, and **Azure-native data engineering architectures**, fostering a culture of technical excellence and continuous improvement.
- Proactively identified and implemented optimizations in existing workflows — including query optimization, storage layout improvements, and pipeline refactoring — that significantly improved performance, reduced costs, and enhanced maintainability.
- Collaborated closely with product, analytics, and leadership teams to ensure that data solutions were aligned with evolving business goals and could adapt quickly to changing market conditions typical of SaaS startups.
- Implemented **data governance frameworks** that enforced schema validation, ownership metadata, and data quality SLAs across Snowflake and BigQuery.
- Established robust monitoring and alerting data pipelines and infrastructure using **Azure Monitor, Log Analytics**, and open-source solutions, ensuring high availability and fast incident response times.

Environment: Python, Scala, PySpark, SQL, Apache Spark, Apache Airflow, Kafka, Kafka Streams, Spark Streaming, Azure Databricks, Azure Data Factory, Azure Synapse Analytics, Azure Data Lake Storage, Azure Blob Storage, Azure Event Hubs, Azure Monitor, Azure DevOps, PowerBI, Terraform, Docker, Kubernetes, Trino/Presto, Git, CI/CD, SaaS analytics, distributed data processing, real-time and batch pipelines, monitoring & alerting tools (CloudWatch, Stackdriver, Prometheus/Grafana), Data governance

Senior Data Engineer

Zendesk, Remote

May 2019 to September 2021

Responsibilities:

- Designed and delivered scalable, cloud-native data pipelines and solutions on **Google Cloud Platform (GCP)** that empowered global strategy and business development teams to make timely, data-driven decisions. Leveraged SaaS analytics platforms to deliver insights at scale in a fast-paced, startup-like environment.

- Built and maintained end-to-end ETL and ELT workflows using **Cloud Composer (Apache Airflow on GCP)**, orchestrating Python and Spark jobs to process large volumes of structured and semi-structured data with high reliability and visibility.
- Architected real-time and batch data processing pipelines leveraging **Apache Kafka, Spark Streaming, Scala, and Python**, enabling ingestion, transformation, and storage of streaming event data in **GCS (Google Cloud Storage)** and **BigQuery** for downstream analytics and reporting.
- Refactored and optimized distributed data applications in Scala and PySpark, **increasing throughput by 45%** and **reducing Dataproc cluster costs by \$15K per quarter** through tuning and workload parallelization.
- Created self-service, reusable **data products and libraries** that allowed business and analytics teams to independently access curated, high-quality datasets while ensuring data integrity and governance compliance.
- Integrated diverse SaaS product data sources (e.g., Salesforce, Stripe, Mixpanel) and internal application logs into unified reporting datasets stored and modeled in **BigQuery**, ensuring consistency and reliability of business metrics across stakeholders.
- Designed and implemented **data warehouse and lakehouse architectures using BigQuery and GCS**, supporting terabytes of data with secure, performant, and automated ingestion pipelines.
- Led the migration of legacy batch ETL processes to real-time, event-driven architectures using Kafka and Spark on **GCP-native infrastructure**, significantly reducing data latency and improving observability.
- Collaborated with cross-functional stakeholders—including product managers, engineers, and analysts—to understand strategic goals and translate them into robust, scalable data engineering solutions aligned with agile delivery cycles.
- Wrote advanced SQL transformations in BigQuery, **reducing dashboard refresh latency by 50%** and enabling faster experimentation for executive and product analytics teams.
- Championed SDLC best practices by implementing **CI/CD pipelines using Cloud Build and GitHub Actions** for data engineering workflows, with unit testing, environment isolation, and automated deployment to production.
- Proactively improved operational excellence by monitoring and optimizing workflows, automating recurring maintenance tasks, and addressing data quality issues through rigorous testing and validation frameworks.
- Mentored junior engineers and data analysts, sharing expertise in **Airflow DAG design, Spark job optimization, and Kafka topic management** to foster a culture of knowledge-sharing and technical excellence.
- Designed solutions to ensure high availability and fault tolerance of critical pipelines, incorporating monitoring, alerting, and retry mechanisms using **Stack driver (Cloud Monitoring)** to handle data quality and system anomalies gracefully.
- Contributed to the development of federated query capabilities using **Trino and BigQuery**, enabling seamless access to disparate data sources for analytics teams without data duplication.
- Designed and deployed **Delta Lake-based architecture** on Databricks to manage CDC and time travel use cases, supporting rollback and auditability for compliance teams.
- Implemented **Great Expectations** across core ELT pipelines to enforce automated data validation rules, reducing production incidents by 65%.
- Actively participated in design and code reviews, bringing deep knowledge of SaaS-specific data challenges, distributed systems, and big data processing paradigms to improve the team's output quality and maintainability.
- Rapidly delivered prototypes for analytics use cases, transforming validated experiments into production-ready pipelines, **accelerating product decision cycles by 3x**.
- Delivered data products visualized through **Tableau**, enabling leadership and product teams to monitor SaaS KPIs, revenue forecasts, and customer engagement trends in near real time.

Environment: GCP (BigQuery, Dataflow, Pub/Sub, Cloud Storage, Dataproc, Cloud Composer, Cloud Build, Stackdriver), Apache Airflow, Apache Kafka, Apache Spark (Streaming & Batch), Scala, Python, PySpark, SQL (analytical & procedural), Jenkins, GitHub Actions, Tableau, Presto/Trino, Great Expectations, SaaS platforms (Salesforce, Stripe, Mixpanel), distributed systems, CI/CD pipelines, monitoring & alerting tools, Data Modeling, Agile.

Senior Data Engineer

Accenture India Pvt. Ltd. Hyderabad, Telangana

August 2016 to February 2019

Responsibilities:

- Designed and implemented robust database architectures by applying normalization (up to 3NF) and selective de-normalization techniques to optimize both OLTP and OLAP workloads; ensured data integrity through well-defined relational constraints and created efficient schema designs to support business needs.
- Tuned SQL queries and optimized database operations by leveraging dynamic SQL, parameterized queries, cursors, and writing modular procedures, functions, and reusable packages to improve application performance and maintainability.
- Built predictive analytics and clustering models using Spark MLlib and Python, identifying process inefficiencies and driving operational improvements in customer behavior analysis.

- Streamlined Hadoop ecosystem operations by creating reusable frameworks and implementing standardized error-handling/logging strategies, improving debugging efficiency by 40%.
- Enabled near real-time system observability by publishing metrics to Splunk and Tableau dashboards, reducing incident detection time by 60%.
- Designed and implemented scalable ETL pipelines to extract, transform, and load data into AWS Data Lake using Spark, Hive, and other AWS-native services, ensuring timely and accurate data availability for downstream consumers.
- Enhanced maintainability and performance of legacy Java systems by implementing unit tests, automated pipelines, and peer reviews; led modernization of applications, resulting in a 25% performance gain.
- Developed visually appealing and highly functional Java Swing desktop interfaces, collaborating closely with front-end teams to ensure seamless integration of back-end services and an intuitive user experience.
- Designed logical and physical data models using DB2, documented relational schemas and data definitions, and optimized database structures for high-performance querying.
- Identified and resolved data inconsistencies through root cause analysis and escalated critical issues promptly to maintain data integrity across systems.
- Assisted in interface testing, developed comprehensive implementation plans, and contributed to smooth deployment of new systems with minimal downtime.
- Facilitated a safe and inclusive environment during team retrospectives, encouraging open communication and ensuring all voices were heard.
- Implemented role-based access control (RBAC) policies on Snowflake and BigQuery to restrict access to PII and finance data, ensuring compliance with internal audit requirements.
- Extracted and engineered features from datasets to train and evaluate classifiers using WEKA, successfully differentiating spam from legitimate messages and improving model accuracy.

Environment : Oracle, DB2, MySQL, Hive, AWS Redshift, Hadoop (HDFS, MapReduce, YARN), Spark (PySpark, Spark SQL, Spark MLlib), Kafka, Apache Flink, Sqoop, Hive, AWS (EMR, S3, Data Lake, Lambda, EC2), Python, Java, Scala, SQL, PL/SQL, HQL, Swing, Eclipse, IntelliJ, Git, WEKA, Tableau, Power BI, Splunk, Excel, Apache Ambari, CI/CD pipelines, Jenkins, Bash scripting, Data Modeling, ETL Design, Data Warehousing, Machine Learning, Predictive Analytics, Caching & JVM Tuning, Unit & Automated Testing, Data Profiling & Quality Assurance.

ETL Developer

Core Technologies, INDIA

November 2014 to July 2016

Responsibilities:

- Managed end-to-end data aspects of projects using Agile Scrum methodology for Software Development Life Cycle.
- Implemented ETL pipelines with Sqoop, Spark, and Hive for data extraction and ingestion into Hadoop Data Lake.
- Utilized advanced SQL with Columnar Databases in AWS for data modeling and analysis.
- Developed Spark programs for batch processing, using Pyspark and Spark-SQL/Streaming for efficient data testing and processing.
- Conducted performance tuning of Spark applications, optimizing Batch Interval time, Parallelism, and memory usage.
- Leveraged Git for version control and collaboration, transforming Hive/SQL queries into Spark transformations using Spark RDDs, Python, and Scala.
- Worked on large datasets for insights generation using Tableau, converted Hive/SQL queries into Spark transformations, and handled ETL processes.
- Utilized Apache NiFi for data movement automation between different Hadoop systems and developed continuous integration and deployment systems using Jenkins and Maven.

Environment: Python, Power BI, AWS Glue, AWS S3, AWS EMR, AWS RDS, Hadoop3.0, AWS, HDFS, SQL, PSQL, Hive2.3, ETL, SQL, Tableau, Spark, Git, Jenkins, Maven.

EDUCATION

Bachelor of Science in Computer Science

2010 to 2014

Osmania University, Hyderabad, India